



BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

PROJECT PROPOSAL

Course No.: EEE 208 (S)

Course Title: Electronic Circuits II Laboratory (Simulation)

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Subsection: A1

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Proposal – 01

Design and Simulation of an Analog PID Controller for DC Motor Speed Regulation

PROJECT PROPOSAL

Design and Simulation of an Analog PID Controller for DC Motor Speed Regulation

EEE 207 — Electronics II — Simulation Platform: OrCAD PSpice 9.2

1. Introduction and Motivation

In modern industrial and consumer electronics, motor speed regulation is typically handled by digital microcontrollers executing control algorithms. However, this digital paradigm often obscures the fundamental physics and continuous-time mathematics that govern control theory.

The motivation for this project is to bridge the gap between abstract control systems and physical electronic components. By designing a purely analog Proportional-Integral-Derivative (PID) controller using Operational Amplifiers (Op-Amps), this project demonstrates how mathematical operations (addition, integration, and differentiation) can be executed instantaneously via hardware.

An analog approach eliminates the quantization errors and sampling delays inherent in digital systems, offering continuous, real-time error correction. Utilizing PSpice allows for the rigorous testing of these analog principles in a safe, highly measurable simulated environment.

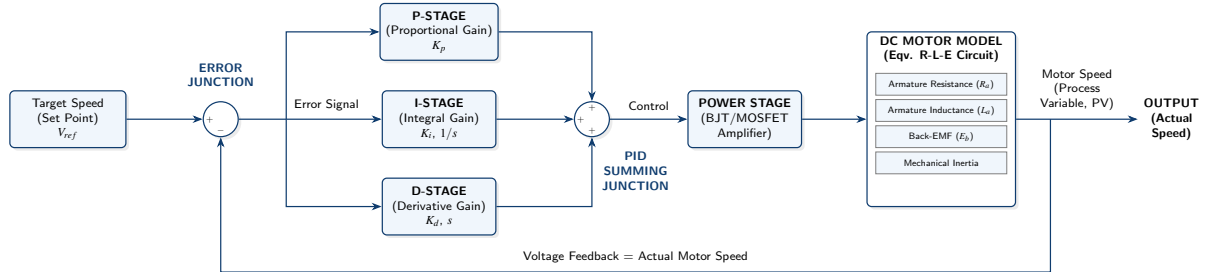
2. Objectives

The primary goal of this project is to successfully simulate a closed-loop motor control system in PSpice. This will be achieved through the following specific objectives:

- **System Modeling:** To construct an accurate electrical equivalent circuit of a permanent magnet DC motor using passive components and dependent sources (representing back-EMF and inertia).
- **Circuit Design:** To design and calculate component values for the error amplifier, proportional, integral, and derivative Op-Amp stages.
- **System Tuning:** To utilize PSpice's parametric sweep capabilities to iteratively tune the controller gains (K_p , K_i , and K_d) to achieve a critically damped system response.
- **Disturbance Analysis:** To prove the robustness of the design by simulating load disturbances and verifying the controller's ability to reject them and maintain the set-point speed.

3. Block Diagram

Below is the functional architecture of the proposed closed-loop system:



4. Methodology

The project will be executed in four distinct phases:

- **Phase 1 - Plant Modeling:** A DC motor does not exist as a standard PSpice component. It will be modeled using an armature resistance (R_a), armature inductance (L_a), and a Voltage-Controlled Voltage Source (VCVS) to represent the back-electromotive force (E_b). A capacitor will be used in parallel to represent the mechanical inertia of the rotor.
- **Phase 2 - Controller Synthesis:** Standard Op-Amp configurations (using the LM741 or ideal Op-Amp models) will be drafted. A difference amplifier will subtract the feedback voltage from the reference voltage. Parallel Op-Amp circuits will be designed where the resistor and capacitor ratios directly dictate the proportional gain, integral time constant, and derivative time constant.
- **Phase 3 - Integration:** The PID outputs will be summed using an inverting summing amplifier, which will then drive a power transistor model to supply sufficient current to the motor equivalent circuit.
- **Phase 4 - Simulation and Optimization:** Transient analysis will be performed in PSpice. A step-voltage will be applied as the reference speed. Parametric sweeps of the proportional feedback resistor will be executed to map out underdamped and overdamped (sluggish) responses, allowing for the precise selection of values that yield critical damping.

5. Equipment and Tools

Category	Item / Specification
Simulation Software	Cadence OrCAD / PSpice
Active Components	Operational Amplifiers (e.g., LM324, LM741, OPAMP models) NPN Power Transistors (e.g., TIP120 or 2N3055)
Passive Components	Resistors, Capacitors, and Inductors
Voltage Sources	Independent DC (VDC), Pulse Voltage (VPULSE for step inputs)
Dependent Sources	Voltage-Controlled Voltage Sources (E part in PSpice)

6. Expected Outcomes

Upon completion of this project, the following deliverables and results are expected:

- **Complete Schematic:** A thoroughly documented PSpice schematic detailing all subsystems (Error calculation, PID stages, Power stage, and Motor model).
- **Transient Response Graphs:** Visual plots demonstrating the motor's speed (voltage) over time responding to a step input.
- **Tuning Proof:** A single graph overlaying multiple parametric sweep iterations, clearly proving the mathematical relationship between the component values and the damping factor of the system.
- **Disturbance Rejection Data:** Simulated evidence showing the controller automatically increasing power to the motor when an artificial load (resistance drop) is introduced to the circuit.